

# Lecture 17 Maximum Likelihood Estimation (MLE)

BIO210 Biostatistics

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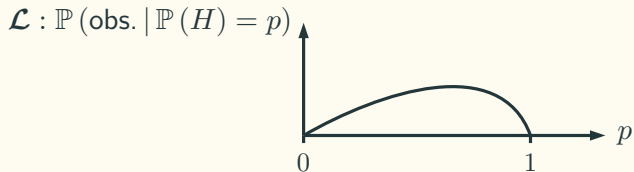
# Intuition over MLE

**Experiment:** A coin, with an unknown  $\mathbb{P}(H) = p$ , was flipped 10 times. The outcome is *HHHTHHHTHH*.

**Question:** What is your best guess for  $p$  ?

**Thinking:** Given the data/observation we have, what values should  $p$  take such that our data/observation is most likely to occur ?

**Aim:** find the value that **maximise our chance of observing the data**, and use that value as our best guess/estimate for  $p$ .



## Maximum Likelihood Estimation (MLE): Example 1

**Example 1:** A coin, with an unknown  $\mathbb{P}(H) = p$ , was flipped 10 times. The outcome is  $HHHTHHHTHH$ . What is the MLE for  $p$ ?

$$\begin{aligned}\mathcal{L}(p; 1110111011) &= \mathbb{P}_{\mathbf{X}}(1110111011; p) = \prod_{i=1}^{10} \mathbb{P}_{\mathbf{X}}(x_i; p) \\ &= \mathbb{P}_{\mathbf{X}}(1; p) \cdot \mathbb{P}_{\mathbf{X}}(1; p) \cdot \mathbb{P}_{\mathbf{X}}(1; p) \cdot \mathbb{P}_{\mathbf{X}}(0; p) \cdot \mathbb{P}_{\mathbf{X}}(1; p) \\ &\quad \cdot \mathbb{P}_{\mathbf{X}}(1; p) \cdot \mathbb{P}_{\mathbf{X}}(1; p) \cdot \mathbb{P}_{\mathbf{X}}(0; p) \cdot \mathbb{P}_{\mathbf{X}}(1; p) \cdot \mathbb{P}_{\mathbf{X}}(1; p) \\ &= p \cdot p \cdot p \cdot (1 - p) \cdot p \cdot p \cdot p \cdot (1 - p) \cdot p \cdot p = p^8(1 - p)^2\end{aligned}$$

$$\text{Let } \frac{d\mathcal{L}}{dp} = 0$$

## Maximum Likelihood Estimation (MLE): Example 2

**Example 2 DNA synthesis errors:** The genetic material is copied and synthesised by DNA polymerase. One high-fidelity DNA polymerase, *Pfu*, originally isolated from the hyperthermophilic archae *Pyrococcus furiosus*, is believed to have very low error rate. Assume the errors generated by *Pfu* follow a Poisson distribution with  $\lambda$  mutations per  $10^6$  base pairs (Mb). We have examined  $n$  newly synthesised DNA fragments and observed that the number of mutations per Mb is  $k_1, k_2, k_3, \dots, k_n$ . What is the MLE for  $\lambda$ ?

$$\mathcal{L}(\lambda; k_1, k_2, \dots, k_n) = \mathbb{P}_{\mathbf{X}}(k_1, k_2, \dots, k_n; \lambda) = \prod_{i=1}^n \frac{\lambda^{k_i}}{k_i!} e^{-\lambda}$$

$$\ell = \ln \mathcal{L} = \sum_{i=1}^n \ln \frac{\lambda^{k_i}}{k_i!} e^{-\lambda}$$

$$\text{Let } \frac{d\ell}{d\lambda} = 0$$

# Estimators of Parameters

- **Parameter space  $\Omega$ :** the set of all possible values of a parameter  $\theta$  or of a vector of parameters  $(\theta_1, \theta_2, \theta_3, \dots, \theta_k)$  is called the parameter space.
  - Bernoulli:  $\theta = p, \Omega = \{p \mid 0 \leq p \leq 1\}$
  - Binomial:  $\theta_1 = n, \theta_2 = p, \Omega = \{(n, p) \mid n = 2, 3, \dots, \text{a finite number}; 0 \leq p \leq 1\}$
  - Poisson:  $\theta = \lambda, \Omega = \{\lambda \mid \lambda \geq 0\}$
  - Normal (Gaussian):  $\theta_1 = \mu, \theta_2 = \sigma^2, \Omega = \{(\mu, \sigma^2) \mid \mu \in \mathbb{R}, \sigma^2 \geq 0\}$
- We refer to an estimator of a parameter  $\theta$  as  $\hat{\theta}$ . An estimator  $\hat{\theta}$  of a parameter  $\theta$  is unbiased if  $\mathbb{E}[\hat{\theta}] = \theta$ . For example,  $\hat{\mu} = \bar{X}$  is an unbiased estimator for  $\mu$ .

# Maximum Likelihood Estimation (MLE) Definition

## Formal definition

Let  $x_1, x_2, x_3, \dots, x_n$  be observations from  $n$  **i.i.d** random variables  $(X_1, X_2, X_3, \dots, X_n)$  drawn from a probability distribution  $f$ , where  $f$  is known to be from a family of distributions that depend on some parameters  $\theta$ . The goal of MLE is to maximise the **likelihood function**:

$$\begin{aligned}\mathcal{L}(\theta; x_1, x_2, x_3, \dots, x_n) &= \mathbb{f}_X \text{ or } \mathbb{P}_X(x_1, x_2, x_3, \dots, x_n; \theta) = \prod_{i=1}^n f(x_i; \theta) \\ &= f(x_1; \theta) \cdot f(x_2; \theta) \cdots f(x_n; \theta)\end{aligned}$$

The log-likelihood function:

$$\ell = \ln \mathcal{L} = \sum_{i=1}^n \ln f(x_i; \theta)$$

## More About Maximum Likelihood Estimation (MLE)

- **Maximum likelihood estimation (MLE)** is a technique used for estimating the parameters of a given distribution, using some observed data.
- Introduced by **R.A. Fisher** in 1912.
- MLE can be used to estimate parameters using a limited sample of the population, by finding particular values so that the observation is **the most likely result to have occurred**.

# Probability vs. Likelihood

$$\mathcal{L}(\theta; x_1, x_2, x_3, \dots, x_n) = f(x_1, x_2, x_3, \dots, x_n; \theta)$$



the likelihood of the parameter(s)  $\theta$   
taking certain values given that a bunch  
of data  $x_1, x_2, \dots, x_n$  are observed.



the joint probability mass/density  
of observing the data  $x_1, x_2, \dots, x_n$   
with model parameter(s)  $\theta$ .

from [Wolfram](#):

**Likelihood** is the hypothetical probability that an event that has already occurred would yield a specific outcome. The concept differs from that of a probability in that a **probability** refers to the occurrence of future events, while a **likelihood** refers to past events with known outcomes.



# Advantages and Disadvantages of MLE

## Advantages:

- Intuitive and straightforward to understand.
- If the model is correctly assumed, the MLE is efficient (meaning small variance or mean squared error).
- Can be extended to do other useful things.

## Disadvantages:

- Relies on assumptions of a model (need to know the PMF/PDF).
- Sometimes difficult or impossible to solve the derivative of  $\mathcal{L}$  or  $\ell$ .
- Sometimes leads to the wrong or biased conclusions